

TSA1_Test_Report

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FIVE TEST CASES

#	Scenario	Input Data	Expected Result	Actual Result	Status	Evidence
1	Normal Approval	Type: STUDENT, Days: 10, Avail: true, Late: 0	Status: APPROVED Late Charge: ₱0.00	Status: APPROVED Late Charge: ₱0.00	PASS	=====
2	Unavailable Reservation	Type: FACULTY, Days: 20, Avail: false, Late: 2	Status: RESERVED Late Charge: ₱4.00	Status: RESERVED Late Charge: ₱4.00	PASS	=====
3	Over-limit Denial	Type: GUEST, Days: 15, Avail: true, Late: 0	Status: DENIED Late Charge: ₱0.00	Status: DENIED Late Charge: ₱0.00	PASS	=====
4	Boundary Value	Type: STUDENT, Days: 14, Avail: true, Late: 3	Status: APPROVED Late Charge: ₱15.00	Status: APPROVED Late Charge: ₱15.00	PASS	=====
5	Summary & Fee	Type: FACULTY, Days: 30, Avail: true, Late: 5	Status: APPROVED Late Charge: ₱10.00	Status: APPROVED Late Charge: ₱10.00	PASS	=====

						<pre> ===== FINAL OPERATIONAL SUMMARY ===== Total Requests Processed: 5 Approved: 3 Reserved: 1 Denied: 1 ===== Request with Greatest Requested Days: -> James (FACULTY) requested 30 days for "Fundamentals of Accountancy" ===== Total Estimated Late Charges: PHP 25.00 ===== BUILD SUCCESS ===== Total time: 05:23 min Finished at: 2026-09-28T20:47:04+08:00 ===== </pre>
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DEBUGGING LOG

Category	Description
Defect Name	Incorrect Rule Precedence and Strict Comparison Operator in classifyRequest()
Defect classification	Logical Error
Defect behavior	1. Exact boundary requests (e.g., 14 days for STUDENT) evaluated to DENIED instead of APPROVED. 2. Unavailable books were marked APPROVED if loan days were within allowed limits.
IDE Debugging Tool Used	NetBeans Debugger
Debugger Steps Taken	1. Placed a Line Breakpoint at the call to classifyRequest() inside the main loop. 2. Inspected variables (requestedDays = 14, maxAllowed = 14, isAvailable = true). 3. Used “Step Into” to watch execution jump past the < conditional directly to DENIED.
Correction Applied	1. Moved if (!isAvailable) check to the top priority position. 2. Replaced requestedDays < maxAllowed with requestedDays <= maxAllowed.
Verification Evidence	Re-ran Test 4 (14 days for STUDENT). Output changed from DENIED to

	APPROVED.